

ASSESSMENT_1

SECTION-A

(Concept Application)

S-1 Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

When selecting hardware for a new office workstation, I would first decide what the computer will mainly be used for. Since it is an office workstation, it should be suitable for applications such as Microsoft Office, web browsing, email and other normal productivity work.

First, I would choose the CPU and check its socket type. Then I would select a motherboard that supports that CPU socket and CPU generation.

Next, I would choose RAM that is supported by the motherboard, such as the correct DDR generation and supported capacity and speed.

For storage, I would choose an SSD because it gives faster startup and application loading. I would also check whether the motherboard supports the required SATA or M.2 connection.

Finally, I would check the motherboard manufacturer's specifications and the CPU, RAM and storage specifications to make sure all the parts are compatible before buying them.

In simple terms, I would check the CPU socket, supported RAM type, storage connection, motherboard specifications and power requirements before assembling the computer.

- Motherboard
- CPU
- RAM
- Storage

S-2 Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

- HDD, SSD and NVMe are different types of computer storage.
- An HDD stores data on spinning magnetic disks. It usually provides a large amount of storage at a lower cost, but it is slower and has moving parts.
- An SSD stores data using flash memory. It has no moving parts and is much faster than an HDD. It also makes the computer start and open applications faster.
- NVMe is a type of SSD that normally uses the PCIe interface. It can provide higher speed and lower access time than a traditional SATA SSD.

For a modern office workstation, I would choose an SSD, preferably an NVMe SSD if the motherboard supports it. Office applications, Windows, web browsers and files will load quickly, which makes the computer more responsive during normal work.

The main reason for choosing an SSD is that it provides better everyday performance than an HDD.

S-3 Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

The GPU, or Graphics Processing Unit, is responsible for processing graphics and other tasks that can be handled by the graphics processor. In a workstation running design software, the GPU helps display and process graphics smoothly.

When the computer runs graphics software for several hours, the CPU and GPU produce heat. The cooling system removes this heat and helps keep the components within a safe operating temperature.

A typical air-cooling system uses a heatsink and fan to move heat away from the component. Good airflow inside the case also helps remove hot air.

Cooling is important because excessive heat can cause the computer to slow down, become unstable or shut down to protect the hardware.

Therefore, the GPU helps handle graphics work, while the cooling system helps maintain safe temperatures and stable performance during long periods of use.

S-4 Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

Hardware compatibility is important because computer components must be able to physically and electrically work together.

For example, the CPU must match the motherboard's CPU socket and supported CPU generation. The RAM must be the correct type supported by the motherboard. Storage must use a connection supported by the motherboard, such as SATA or M.2.

If incompatible components are used, the computer may not start. A component may also fail to be detected, or the system may become unstable.

Another possible problem is that the power supply may not provide enough power or the correct connectors for the components.

Therefore, I would check the specifications of every component before assembling the PC. This helps prevent installation problems and possible hardware damage.

What if we ignore compatibility:

- CPU won't fit the motherboard socket.
- RAM type is not supported.
- Storage connector is not available.
- PSU does not provide the required connectors/power.
- GPU will not physically fit the case.
- BIOS may not support a particular CPU.

S-5 Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

I would first estimate the total power requirement of the CPU, GPU, storage devices and other components. I would then select a reliable PSU with enough wattage and the correct connectors. I would also leave some extra power capacity instead of selecting a PSU that is exactly at the calculated requirement.

Before connecting the PSU, I would make sure the power is disconnected. I would connect the 24-pin ATX cable to the motherboard and the CPU power cable to the CPU power connector.

If the graphics card requires additional power, I would connect the correct PCIe power cable. SATA storage devices would receive power through SATA power connectors.

Some connectors required are:-

MOtherboard- 24-pin ATX

CPU- 4/8 pin EPS CPU power

GPU- PCIe power cable(if required)

SATA storage- SATA power connector

Before the first boot, I would check that all connectors are firmly connected, there are no loose screws or cables, and the components are correctly installed.

Only after completing these checks would I connect the PSU to mains power and switch on the system.

S-6 Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

POST stands for Power-On Self-Test. It is performed when the computer starts and checks important hardware components.

To confirm that POST is successful, I would switch on the computer and check that the system powers on, the fans operate and a display appears. I would check that the BIOS or manufacturer's startup screen appears without a hardware error.

If the computer does not pass POST, I would first switch off the power and check the basic connections. I would check the PSU and motherboard power cables, RAM seating, CPU installation and GPU/display connection.

I would also check for any beep codes, diagnostic LEDs or error messages from the motherboard. I would correct the suspected hardware problem and test the system again.

I would make changes one at a time so that I can identify the actual cause of the problem.

SECTION-B

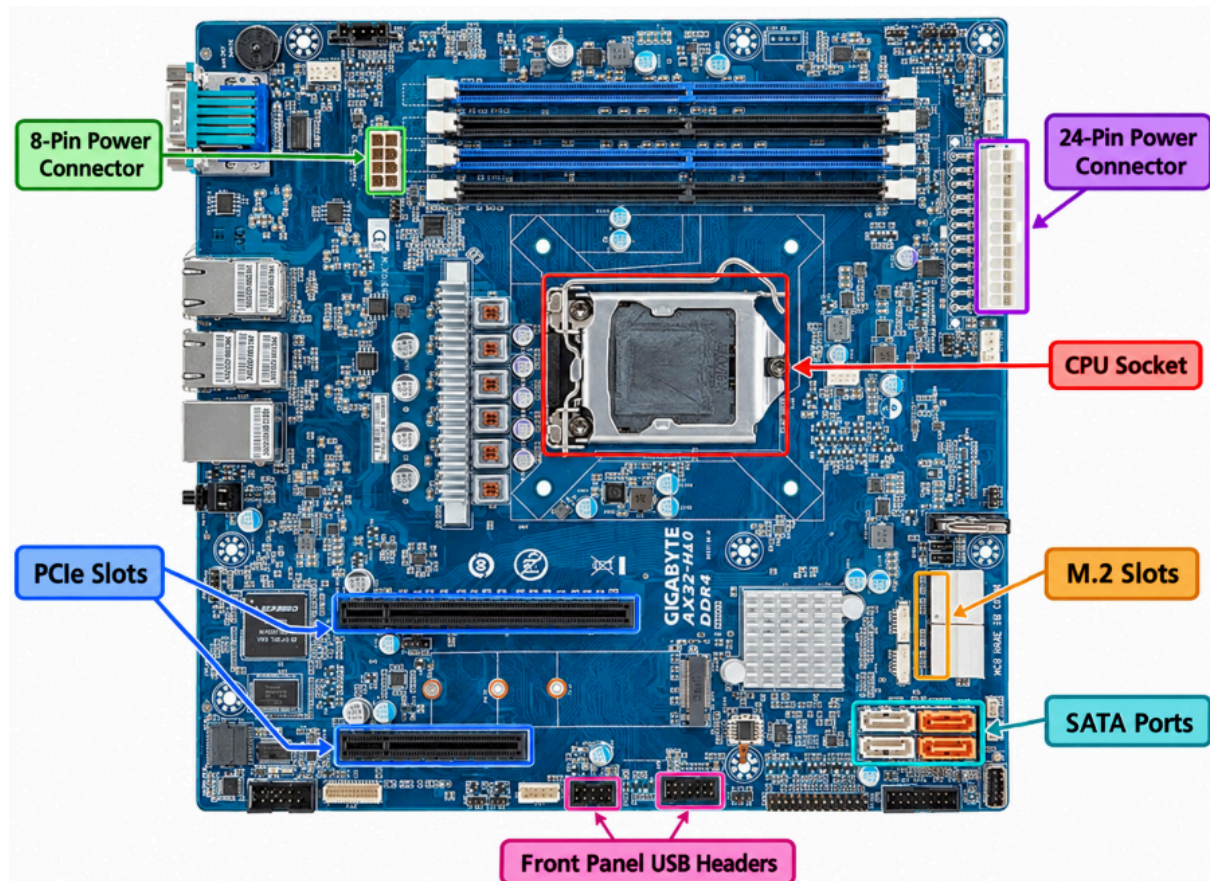
(Practical Coding Task)

TASK-1 Component Identification and Function.

Model	Function	Key-specification
CPU	The CPU processes instructions and performs calculations needed by the operating system and applications.	Intel Core i5-1155G7, 11th Gen
RAM	RAM temporarily stores data and instructions that the CPU is currently using. It helps applications run smoothly.	8GB
GPU	he GPU processes graphics and sends the image to the display.	Intel Iris Xe Graphics — integrated
Motherboard	The motherboard is the main circuit board. It connects the CPU, RAM, storage, GPU and other hardware so they can communicate with each other.	
Storage	Storage permanently keeps the operating system, applications and files	477 GB total capacity shown in Windows (NVMe)
Cooling system	The cooling system removes heat from components and helps keep temperatures within a safe range.	Laptop fan + heatsink/cooling system

TASK-2 Motherboard Ports & Socket Identification.

Part	Purpose
CPU Socket	Holds and connects the CPU to the motherboard.
RAM/DIMM Slots	Used to install desktop RAM modules.
PCIe x16 Slot	Used mainly for a dedicated graphics card.
PCIe Expansion Slots	Used for additional expansion cards.
M.2 Slot	Used for compatible M.2 storage devices, including NVMe SSDs.
SATA Ports	Used to connect SATA storage devices.
24-pin ATX Connector	Provides the main power connection to the motherboard.
CPU/EPS Power Connector	Provides power to the CPU area.
USB Headers	Connect internal USB ports from the PC case to the motherboard.



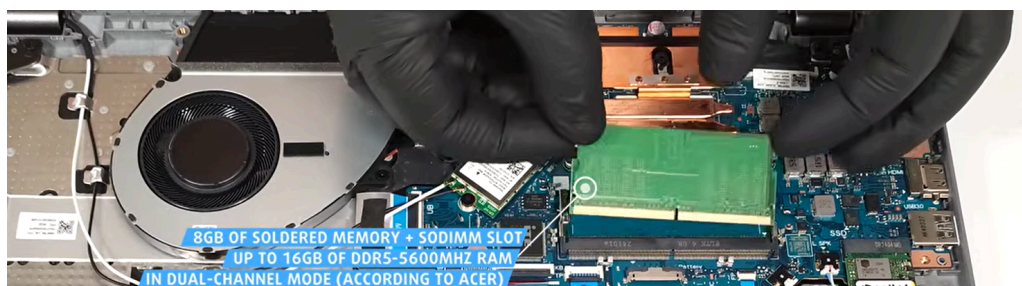
TASK-3 RAM and Storage information.

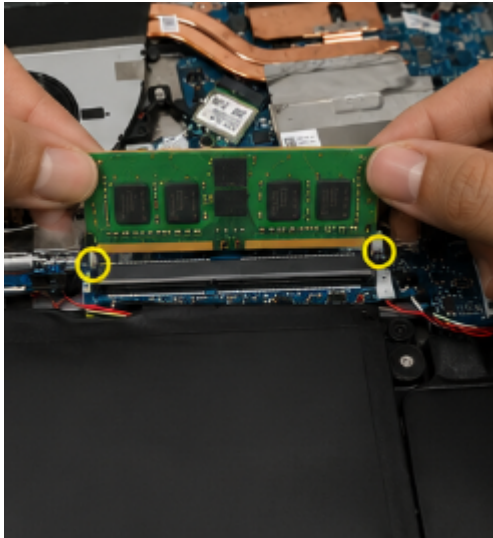
Practical Method:

This task was completed using a PC hardware simulator/reference diagrams instead of physically opening my laptop.

RAM Installation

1. Shut down the computer and disconnect power.
2. Locate the RAM slot.
3. Open the side clips and remove the RAM module carefully.
4. Align the RAM notch with the slot and press it down until the clips lock.
5. Make sure the RAM is seated correctly.





Storage Installation

M.2/NVMe SSD:

- Remove the retaining screw.
- Carefully remove the SSD.
- Insert it back into the M.2 slot and secure it with the screw.



Safety Precautions

- Disconnect power before working.
- Use ESD protection.
- Hold components by their edges.
- Do not touch the gold contacts.

Verification

After installation, check BIOS/UEFI and Windows to confirm that the RAM and storage device are detected. (I have watched youtube video for this task and all ss are from youtube videos).

TASK-4 Full Desktop PC Assembly

Step 1 — Prepare the Work Area

I would use a clean, dry and well-lit workspace. I would disconnect power while installing components and take ESD precautions.

Step 2 — Install the CPU

I would identify the correct CPU socket on the motherboard. I would align the CPU correctly with the socket markings and place it carefully into position without forcing it.

Step 3 — Install the CPU Cooler

I would install the compatible CPU cooler according to its mounting instructions. If thermal paste is required, I would apply the correct amount and make sure the cooler is securely attached.

Step 4 — Install RAM

I would identify the correct RAM slots, align the notch on the RAM module and press the module into the slot until the retaining clips lock.

Step 5 — Install the Motherboard

I would place the motherboard into the PC case and secure it using the correct screws and mounting points.

Step 6 — Install Storage

I would install the storage device in its correct location. A SATA drive requires a SATA data connection and SATA power. An M.2/NVMe drive is installed into a compatible M.2 slot.

Step 7 — Install the GPU

If the system uses a dedicated graphics card, I would insert it into the appropriate PCIe x16 slot and secure it to the case. I would connect the required PCIe power cable if the GPU needs additional power.

Step 8 — Install the PSU

I would install the PSU in the case and secure it properly.

Step 9 — Connect Power

I would connect the 24-pin motherboard power cable and CPU power cable. I would also connect the required GPU and storage power cables.

Step 10 — Final Inspection

Before switching on the computer, I would check that all components are seated correctly and all required cables are connected. I would also check for loose screws, loose cables or anything that could cause a short circuit.

Step 11 — First Boot and POST

I would connect the monitor, keyboard and power cable and switch on the computer. I would check that the system powers on and displays the BIOS/startup screen without a POST error.

SECTION-C

(Mini Capstone Project)

Part 1: First Three Hardware Checks

If a newly assembled PC does not boot, I would first check the following three areas:

1. PSU and Power Connections

I would check whether the PSU is switched on and connected correctly. I would also check the 24-pin motherboard power connector and CPU power connector. A missing power connection can prevent the system from starting.

2. RAM

I would check whether the RAM is installed in the correct slot and seated properly. Incorrectly installed RAM can cause the computer to fail POST or produce a memory error.

3. Motherboard and CPU

I would check the motherboard for correct power connections and make sure the CPU is correctly installed and supported by the motherboard. I would also check any motherboard diagnostic LEDs or error codes.

These checks are useful because they cover the basic power, memory and main processing hardware needed for the system to start.

Part 2: Motherboard and CPU Compatibility

Motherboard and CPU compatibility is important because the CPU must physically fit the motherboard socket and the motherboard must support that CPU.

I would first check the exact CPU model and the motherboard model. Then I would check the manufacturer's specifications for the CPU socket, supported processor families and BIOS support.

If an incompatible CPU is installed, the computer may fail to boot or the CPU may not be detected.

For the system I inspected, I would record the exact motherboard and CPU models and compare them with the manufacturer's compatibility information before assembly.

Part 3: RAM and Storage Verification

After installing the RAM, I would start the computer and enter the BIOS/UEFI if necessary. I would check whether the expected amount of RAM is detected.

I would also check whether the storage device appears in BIOS/UEFI. This confirms that the motherboard can detect the installed storage device.

After Windows starts, I would check the available memory and verify that the storage device is visible in the operating system.

If either component is missing, I would switch off the system and check the physical installation, connectors and compatibility before testing again.

Part 4: Hardware Installation Checklist

- ☐ Check motherboard and CPU compatibility before installation.
- ☐ Check that the RAM type is supported by the motherboard.
- ☐ Install RAM correctly and make sure the retaining clips are locked.
- ☐ Check that the storage device uses a supported SATA or M.2 connection.
- ☐ Connect the 24-pin motherboard power cable and CPU power cable correctly.
- ☐ Connect GPU and storage power cables where required.
- ☐ Perform a final physical inspection before powering on the system.
- ☐ Check BIOS/UEFI for RAM and storage detection after installation.
- ☐ Confirm that the system completes POST successfully.

Part 5: SSD vs HDD for Office Workstations

For the ABC Solutions computer lab, I would use SSD storage instead of HDD storage for the main system drive.

An SSD has no moving mechanical parts and provides faster access to data than a traditional HDD. This helps Windows start faster and allows applications to open more quickly.

In an office environment, users may frequently open documents, web browsers, productivity applications and other files. Faster storage makes these everyday tasks more responsive.

An SSD can also reduce noise because it does not use spinning disks.

Therefore, an SSD is a suitable choice for modern office workstations because it provides faster everyday performance and a better user experience.

SECTION-D

(AI- Argumented Learning)

Task 1 – Build with AI(Chatgpt)

AI Prompt Used

“Create a step-by-step troubleshooting checklist for a desktop PC that fails to boot after assembly. Include the motherboard, RAM, storage and PSU as possible failure points. Arrange the checks in the correct order and make it easy to follow.”

Original AI Checklist

1. Check that the PSU is switched on and connected properly.
2. Check the motherboard power cables.
3. Check whether the RAM is installed correctly.
4. Check the CPU and motherboard compatibility.
5. Check whether the storage device is connected correctly.
6. Check the GPU and other components.
7. Start the PC and check for POST/BIOS errors.

Task 2 – Test & Debug Without AI

The original checklist was useful, but I found that it could be improved by making the steps more focused on the basic power and POST checks first.

Corrected Checklist

1. Check the PSU and power connections.
2. Check the motherboard power cables.
3. Check that the CPU is installed and compatible with the motherboard.
4. Reseat the RAM and check whether it is detected.
5. Check the GPU connection if a dedicated GPU is installed.
6. Check the storage connection.
7. Start the PC and check POST/BIOS for error messages.

What I Changed and Why

I kept the main steps from the AI answer but improved their order. Power and motherboard connections should be checked first because the computer cannot start without them. I also placed CPU, RAM and GPU checks before storage because these components are more important for getting the system through POST.

This project showed me that AI can create a useful troubleshooting checklist, but the answer should still be reviewed. Checking and correcting the AI's steps makes the troubleshooting process clearer and more logical.